

Software Re-engineering (SE4001) Final Exam

Date: May 29th 2024

Course Instructor(s)

Dr. Syed Muazzam Ali Shah

Total Time (Hrs):

03

Total Marks:

100

Total Questions:

04

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

CLO # 4: Investigate and design complex reengineering and reverse engineering problems

Q1: Answer the following questions

[55 marks]

A. Data Flow Analysis:

[15 marks]

- a. Consider the given data flow diagram (**Fig. 01**) and specify whether or not **d** reaches **u**, and also identify the path through which **d** reaches **u**. Provide valid justification for your answer. (7 marks)

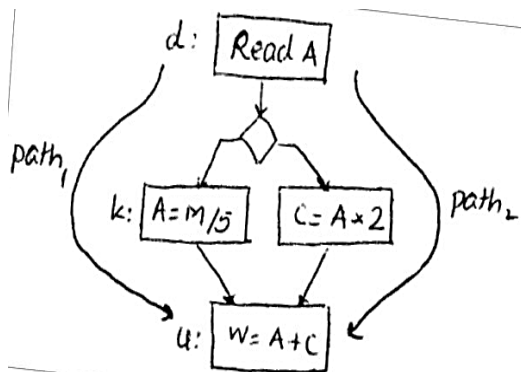


Fig. 01: data flow diagram

- b. Consider the given data flow diagram (**Fig. 02**) and identify the following with valid justification. (8 marks)

1. The assignment **y = 1** reaches **P1**
2. The assignment **y = 1** reaches **P2**
3. The assignment **y = x * y** reaches **P1**

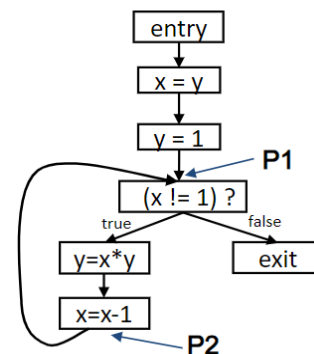


Fig. 02: data flow diagram

Solution (Part a):

d reaches **u** along path 2 & **d** does not reach **u** along path 1 Since there exists a path from **d** to **u** along which **d** is not killed (i.e., path 2), **d** reaches **u**.

Solution (Part b):

1. The assignment **y = 1** reaches **P1** ☒
2. The assignment **y = 1** reaches **P2** ☐
3. The assignment **y = x * y** reaches **P1** ☒

B. Control Flow Analysis & Syntactic Analysis

[15 marks]

- a. Draw a control flow diagram for the given source code fragment: (8 marks)

```
x = 5;
while (x != 0) {
    y = x;
    x = x - 1;
    while (y != 0) {
        y = y - 1;
    }
}
```

- b. Consider the following expression and construct an abstract syntax tree such that the leaf nodes represent the numbers while the intermediate nodes represent the operators:

• $4 + 2 * 10 + 3 * (5 + 1)$

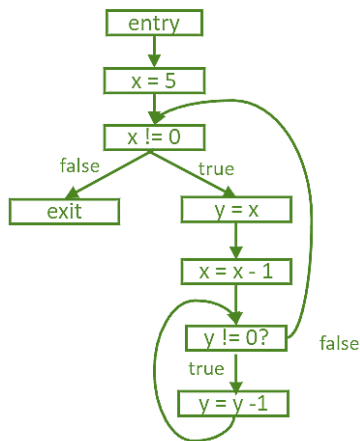
***Note:** The tree must accurately represent the given expression

(7 marks)

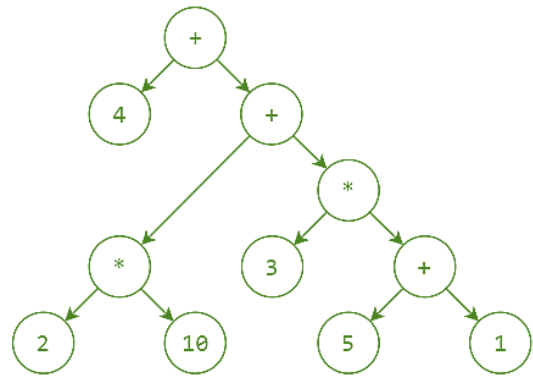
National University of Computer and Emerging Sciences

Karachi Campus

Solution (Part a):



Solution (Part b):



C. Program Slicing

[10 marks]

Using the source code below, perform program slicing using the slicing criteria $\langle S, V \rangle$, where S is the statement number, and V is the variable. For the given source code, $S = 12$, and $V = \text{average}$. Explicitly specify the sliced program after applying the slicing criteria, and also justify your answer.

```

1. DataInputStream d = new
   DataInputStream (System.in);
2. terminate_var=
   Integer.parseInt(d.readLine());
3. product=1;
4. sum=1;
5. for(count=1; count<=terminate_var;
   count++){
6. sum=sum+count;

```

```

7. product=product*count;
8. }
9. average=(sum-1)/terminate_var;
10. System.out.println("The Sum is : "+sum);
11. System.out.println("The Product is : " +
   product);
12. System.out.println("The Average is : "
   +average);

```

Solution:

In the below program the code that is highlighted in green color is the desired program slice that fulfills the given slicing criteria. Here backward slicing technique is the most appropriate and suitable.

```

1. DataInputStream d = new
   DataInputStream (System.in);
2. terminate_var=
   Integer.parseInt(d.readLine());
3. product=1;
4. sum=1;
5. for(count=1; count<=terminate_var;
   count++){
6. sum=sum+count;

```

```

7. product=product*count;
8. }
9. average=(sum-1)/terminate_var;
10. System.out.println("The Sum is :
   "+sum);
11. System.out.println("The Product is :
   " + product);
12. System.out.println("The Average is :
   " +average);

```

D. Lexical Analysis & Data Flow Analysis

[15 marks]

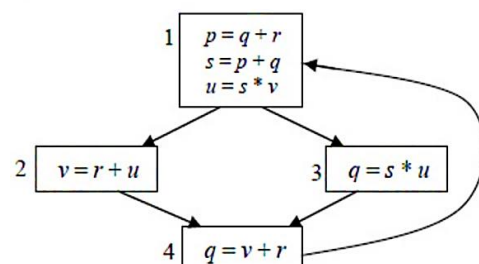
a. Perform the lexical analysis for the given source code fragment: (8 marks)

```

! start of declarations
real a;
real baboon;
int x, y;
! end of declarations
x := 300;
printi(x);
y := 7 - x;
a := -0.12 + +12.34 + 12 / 3 * 5 - 5;
baboon := a * y
printi(5);
printr(baboon);

```

b. Consider the given data flow diagram and determine the variables that are live at each block; here 1, 2, 3, and 4 signify Block 1, 2, 3, and 4, respectively. (7 marks)



National University of Computer and Emerging Sciences

Karachi Campus

***Note:** Determine the total number of tokens, along with the categories of the tokens.

Fig. 03: data flow diagram

Solution (Part a):

#	token	category	#	token	category	#	token	category
1	real	keyword	21	y	identifier	41	baboon	identifier
2	a	identifier	22	:=	operator	42	:=	operator
3	;	punctuation	23	7	constant	43	a	identifier
4	real	keyword	24	-	operator	44	*	operator
5	baboon	identifier	25	x	identifier	45	y	identifier
6	;	punctuation	26	;	punctuation	46	;	operator
7	int	keyword	27	a	identifier	47	printi	keyword
8	x	identifier	28	:=	operator	48	(	lparn
9	,	punctuation	29	-0.12	neg constant	49	5	constant
10	y	identifier	30	+	operator	50	)	rparn
11	;	punctuation	31	+12.34	pos constant	51	;	operator
12	x	identifier	32	+	operator	25	printr	keyword
13	:=	operator	33	12	constant	53	(	lparn
14	300	constant	34	/	operator	54	baboon	identifier
15	;	punctuation	35	3	constant	55	)	rpran
16	printi	keyword	36	*	operator	56	;	punctuation
17	(	lparn	37	5	constant			
18	x	identifier	38	-	operator			
19	)	rparn	39	5	constant			
20	;	punctuation	40	;	punctuation			

Solution (Part b):

Block 1: variables **r**, and **v** are live

Block 2: variables **r** and **u** are live

Block 3: variables **s**, **u** are live

Block 4: variables **v** and **r** are live

CLO # 2: Demonstrate & utilize reengineering techniques to maintain and modify software systems

Q2: Generate the sequence diagram using the following source code.....

[10 marks]

```

import revuml.test.OOConcept.inheritance.Animal;
import revuml.test.OOConcept.inheritance.Cat;
public class REVUMLTest {
    public static void main(String[] args)
    {
        int a, b;
        String name = "Kitty";
        a = 10;
        b = 5;
        Animal A = new Animal();
        A.setName (name) ;
    }
}

```

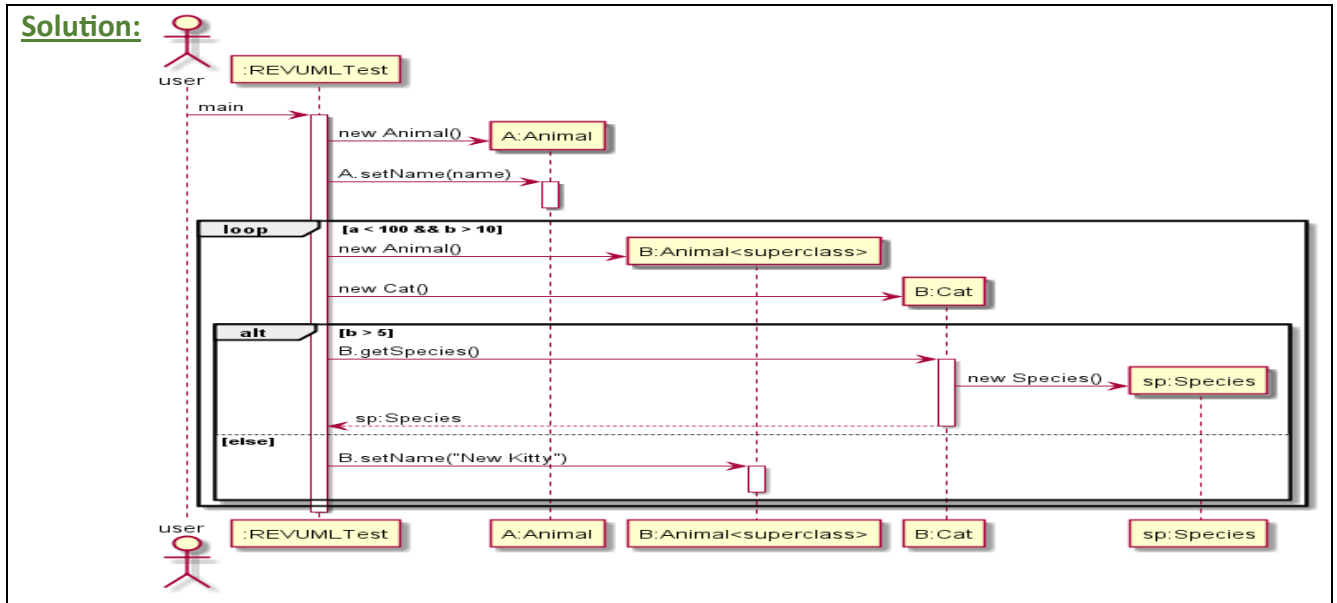
```

while (a<100 && b>10){
    Cat B = new Cat();
    if (b > 5) {
        B.getSpecies();
    } else {
        B.setName ("New
        Kitty");
    }
}
}
}

```

National University of Computer and Emerging Sciences

Karachi Campus



CLO # 1: Define the concepts and techniques of software reengineering

Q3: Answer the following questions

[25 marks]

1. Discuss the three applications of program slicing.

Solution:

Debugging:

Focus on parts of program relevant for a bug. If you are interested in a particular misbehavior of your program i.e, some bug, then slicing helps you to identify which parts of the program influence what you are seeing. You know that at a particular point in the program some variable has the wrong value then the question you are typically asking is which parts of the program influence the value of that variable that I'm seeing here and slicing can help you to answer that question by basically focusing your attention on those parts of the program that are relevant for this bug.

Program understanding:

Which statements influence this statement?

You may just want to know which statements are actually influencing some other statements and the values that they are computed on and slicing can help you by doing. This you can start from one statement and then either slice backward or forward and see how this statement gets influenced or influences future statements.

Change impact analysis:

Which parts of a program are affected by a change? What should be retested? When a program is evolving then the code is changing then at some point you want to know which parts of the code are affected by a particular change. For example, this is interesting to know if you want to decide which parts of a program should be retested or may be tested at all after a code change and because typically you do not have to re-test everything after you change something but only those parts of the program that are affected by a change and which parts of a program are affected by a change that's something you can compute with the help of slicing.

Parallelization:

Determine parts of program that can be computed independently of each other. If you have a program that is sequential but you would like to run it in parallel to speed up the execution then you need some way to determine which parts of the program can be computed independently of each other and because slicing tells you about the dependencies of different statements and parts of your program it can also help you to basically compute individual slices that are independent of each other and then you can run these slices in parallel because they anyway do not influence each other.

2. Specify the six objectives of reverse engineering as documented by the IEEE standard for Software Maintenance.

Solution:

- Partition source code into units.
- Describe the meanings of those units and identify the functional units.
- Create the input and output schematics of the units identified before.
- Describe the connected units.
- Describe the system application.
- Create an internal structure of the system.

3. What does it mean to preserve the behavior of the software and maintain the consistency of the software artifacts during the refactoring process? Explain.

Solution:

National University of Computer and Emerging Sciences

Karachi Campus

- **Ensure that Refactoring Preserves the Behavior of the Software:**

Ideally, the behavior of a program after refactoring should be the same as the behavior before refactoring. Program behavior simply referred to input–output behavior. In other words, for the same set of input values, the programs before refactoring and after refactoring were desired to produce the same output values. However, in many applications preservation of input–output behavior alone is not enough, because preservation of temporal constraints and non-functional requirements of the program may be key to the success of refactoring.

- **Maintain Consistency of Software Artifacts:**

A software system is described by many artifacts at different levels of abstractions. Those artifacts include requirements documents, design documents, source code, and test suites. If one kind of artifact is changed, then it is important to change some or all of the other artifacts so that consistency is maintained across the artifacts.

4. What do external software value and internal software value mean in the context of software restructuring?

Solution:

A higher level goal of software restructuring is to increase the software value as explained in the following:

External software value: This is the value of the software as perceived by the customers. For example, a software with faults may fail to satisfy the business needs of the customers. Such software may be seen by the customers to have less value.

Internal software value: This represents the cost saving due to three factors: 1. Maintenance cost saving due to a good structure of the software; 2. Cost saving due to potential reuse of components of a software with good structure; 3. Cost saving due to extended use of a software.

5. State two merits and two demerits of the incremental approach of software re-engineering.

Solution:

- **Advantages:**

- Locating errors becomes easier (newly added components).
- It becomes easy for the customer to notice progress, because interim versions are released.
- Lower risk due to code can be identified and monitored.

- **Disadvantages:**

- Needs long time and careful version controls.
- Even if there is a need, the entire architecture of the system cannot be changed.

CLO # 3: Examine maintenance related problems associated with object-oriented software systems

Q4: Identify the design pattern for the given case study, draw UML diagram, write the source code and show the output of the code. [10 marks]

Assume that you are solution architect and about to design a java-based software for multiple media images reading and then encoding the images later on. You are focusing on writing different image readers (gif, tiff, jpeg, and bmp) as separate classes, which will implement the same ImageReader interface. Each concrete class BMPReader, GifReader, JpegReader, TiffReader will have a readFile() method whose signature will also be present in the interface. Each reader will be responsible for reading a particular type of image. Your client code will not be responsible for creating any reader object type, instead are interested to decouple this functionality from your client main() code (PatternDemo). What kind of design pattern would you be implementing in this application? And why? Write complete application code with the main() method in the client / driver code to simulate client asking to create different image reader objects from your pattern. Also show sample output of the code that you intend to develop. In the driver code try to read all the following type of files tiff, gif, jpeg, bmp and show the output.

Solution:

Design Pattern: Factory Design Pattern.

```
public interface Image {
    public void readFile();
}

public class BMPReader implements Image {
    public void readFile() {
        System.out.println("Reading a BMP file");
    }
}

public class GifReader implements Image {
    public void readFile() {
        System.out.println("Reading a Gif file");
    }
}

public class JpegReader implements Image {
    public void readFile() {
```

National University of Computer and Emerging Sciences

Karachi Campus

```
        System.out.println("Reading a Jpeg file");
    }
}
public class TiffReader implements Image {
    public void readFile() {
        System.out.println("Readinig a Tiff file");
    }
}
public class ImageFactory {
    public Image getImage(String Shapetype){

        if(Shapetype.equalsIgnoreCase("JPEG")){
            return new JpegReader();
        }
        else if(Shapetype.equalsIgnoreCase("BMP")){
            return new BMPReader();
        }
        else if(Shapetype.equalsIgnoreCase("GIF")){
            return new GifReader();
        }
        else if(Shapetype.equalsIgnoreCase("TIFF")){
            return new TiffReader();
        }
        else
            return null;
    }
}
public class FactoryPatternDemo {
    public static void main(String[] args) {
        try {
            ImageFactory ifactory =new ImageFactory();
            ifactory.getImage("jpeg").readFile();
            ifactory.getImage("bmp").readFile();
            ifactory.getImage("tiff").readFile();
            ifactory.getImage("gif").readFile();
            ifactory.getImage("psd").readFile();
        } catch (Exception e) {
            System.out.println("exception is :"+e.getMessage());
        }
    }
}
```

Output:

Reading a Jpeg file

Reading a BMP file

Reading a Tiff file

Reading a Gif file

exception is :Cannot invoke "Image.readFile()" because the return value of
"ImageFactory.getImage(String)" is null